

TEZPUR UNIVERSITY

Autumn Semester End Examination, December, 2017

CO509: Data Mining and Data Warehousing

Full Marks: 60

Time: 2 hours

1. Fill up the blanks: 1×5=5
 - (a) The leaf nodes of a decision tree stores ____.
 - (b) Density based clusters with widely varying density can be detected with the ____ algorithm.
 - (c) In a 4-dimensional data cube there are ____ cuboids.
 - (d) A dimension table may be chaired by two fact tables in the ____ schema of data warehousing.
 - (e) A ____ allows data to be modelled and viewed in multiple dimensions.
2. State true or false: 1×5=5
 - (a) The *Slice* operation deals with merging cells of all but one dimension.
 - (b) The *Silhouette* measure combines the notion of cluster compactness and separation.
 - (c) Full materialization of a data cube precomputes all of the cuboids.
 - (d) The accuracy of a decision tree is calculated using the *Gini* index.
 - (e) Information gain is an attribute selection measure for constructing a decision tree.
3. What are the major tasks that need to be performed in order to evaluate whether the clustering result are good or not? 3
4. Differentiate between: *OLTP* and *OLAP* 3
5. (a) Define an association rule. Discuss the importance of discovering association rules. 4
 - (b) What steps are followed by the *Apriori* algorithm to find frequent itemsets. 4
 - (c) Mention some drawbacks of the *Apriori* algorithm. 2
6. (a) Who needs a data warehouse and why? 2
 - (b) Present 3-tier architecture of a data warehouse. 3
 - (c) What is a multi-dimensional data model? Explain the *OLAP* server in detail. 5
 - (d) State the different applications of a data warehouse. 2
7. (a) Explain the technique of classification by decision tree induction. 5
 - (b) Discuss the problem of overfitting and underfitting a decision tree. 3
 - (c) What are different methods commonly used to evaluate accuracy of a classifier. 2
8. Given the following transaction dataset:

TID	Items Purchased
1	a, b, c
2	a, d, e
3	a, c
4	b, e
5	b, c
6	a, c, d, e
7	b, c, d

- (a) Find all frequent itemsets using the *Apriori* algorithm. Show the computation steps.
Take minimum support (*minsup*) = 20%. 5
- (b) Find two strong association rules from any one of the frequent 3-itemsets you have found.
Take minimum confidence (*minconf*) = 80%. 3
- (c) Construct the *FP*-tree for the dataset given above. 4